



# General purpose PLL

2026 May 27

## Super Long Wave

Yutaka Kotani (team leader) [yutaka.kotani.in@gmail.com](mailto:yutaka.kotani.in@gmail.com)

Ashutosh Chakravarty

Cheng Fei

# Agenda

1 Project information

2 Team background

3 Specification

4 Block diagram

5 Circuit design

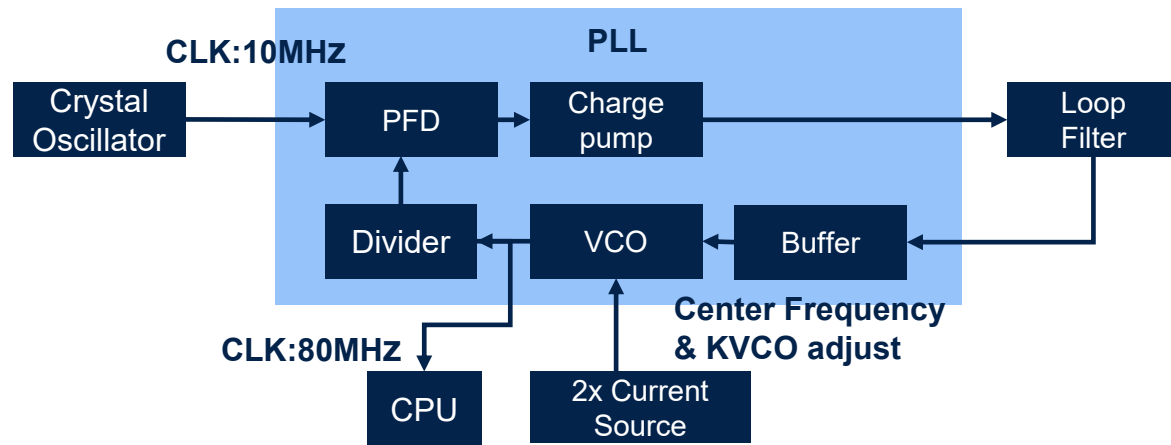
6 Layout

7 Evaluation

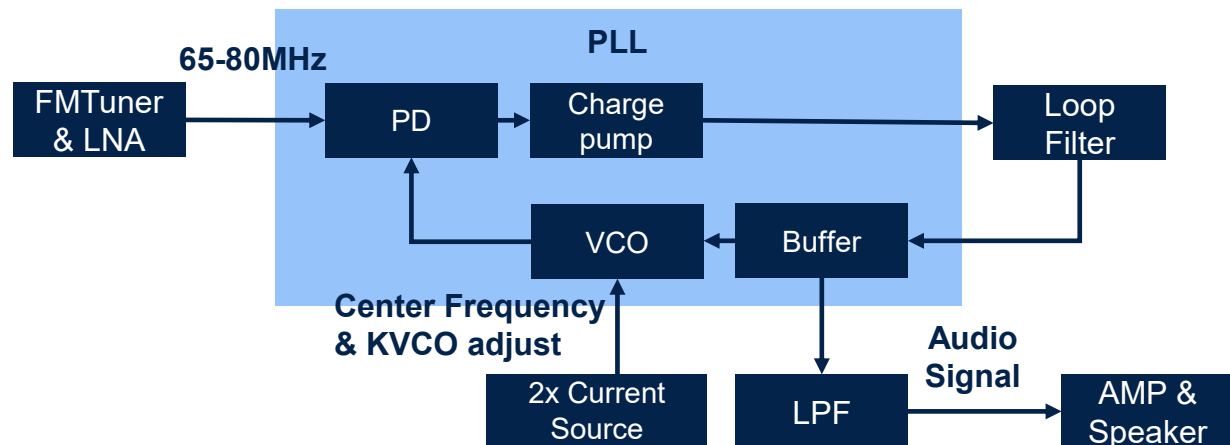
8 Schedule

# Project information

## Clock multiplier mode



## FM demodulator mode



- Goal
  - Creating general purpose PLL
  - Creating a part of FM Radio component
- Design
  - PLL has two mode
  - Clock multiplier mode
  - FM demodulator mode
- Application
  - Digital clock generator & FM Radio
- Reference
  - TI CD4046B Application Report SCHA002A

# Team background

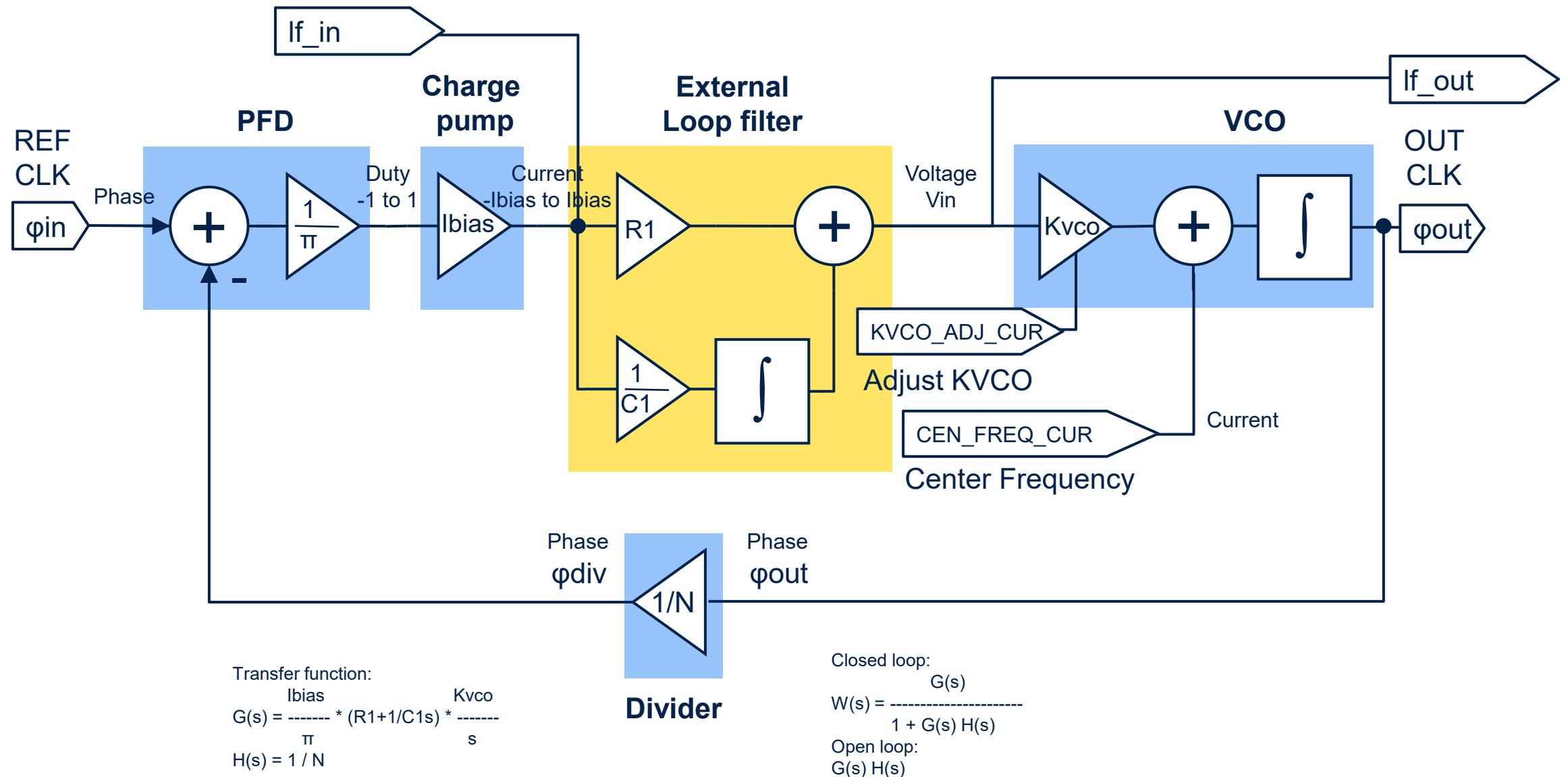
|                     | <b>Yutaka Kotani</b>                                | <b>Ashutosh Chakravarty</b>                     | <b>Cheng Fei</b>        |
|---------------------|-----------------------------------------------------|-------------------------------------------------|-------------------------|
| Discode name        | Yutaka KOTANI                                       | Teddy                                           | buttercutter            |
| Academic experience | Bachelor of Computer Science                        | Bachelors in EE                                 | EE                      |
| Work experience     | Field Application engineer in Semiconductor company | Analog Design Engineer with tape out experience | Digital Design Engineer |

# Specification

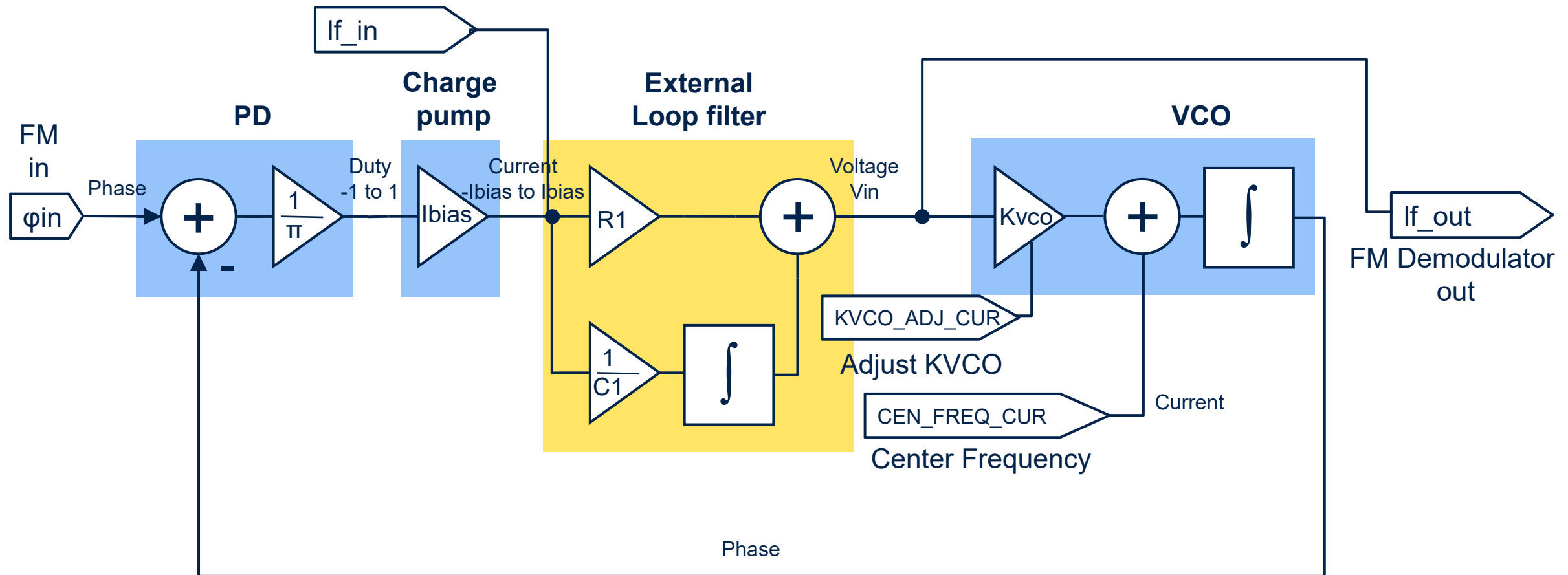
| Target Spec             | Min    | Typ | Max  |
|-------------------------|--------|-----|------|
| Pin count               | 11     |     |      |
| Estimated Area          | 999um2 |     |      |
| Vdd voltage [V]         | 3      | 3.3 | 3.6  |
| pll_in voltage [V]      | 0.8    | -   | 2.5  |
| pll_out voltage [V]     | 0      |     | vdd  |
| Clock multiplier mode   | Min    | Typ | Max  |
| pll_in frequency [MHz]  | 9.9    | 10  | 10.1 |
| pll_out frequency [MHz] | 79.9   | 80  | 80.1 |
| Divider ratio           |        | 1/8 |      |
| Cen_freq_cur [uA]       | 90     | 100 | 110  |
| FM demodulator mode     | Min    | Typ | Max  |
| pll_in frequency [MHz]  | 76     | -   | 90   |
| dmod_out voltage [V]    | 0      | -   | vdd  |
| Cen_freq_cur [uA]       | 10     | -   | 110  |

| Pin | Pin name     | Notes                                               |
|-----|--------------|-----------------------------------------------------|
| 1   | VDD          | 3.3V                                                |
| 2   | GND          |                                                     |
| 3   | PLL_IN       | Reference clk input                                 |
| 4   | PLL_OUT      | Clock multiplier output                             |
| 5   | CEN_FREQ_CUR | Center frequency bias current                       |
| 6   | KVCO_ADJ_CUR | kvco adjust bias current                            |
| 7   | MODE_SEL     | Hi: Use FM demodulator<br>Low: Use Clock multiplier |
| 8   | EXT_FIL_OUT  | To Band pass filter 10.7MHz                         |
| 9   | EXT_FIL_IN   | From Band pass filter 10.7MHz                       |
| 10  | LF_IN        | External Loop filter input                          |
| 11  | LF_OUT       | External Loop filter output                         |

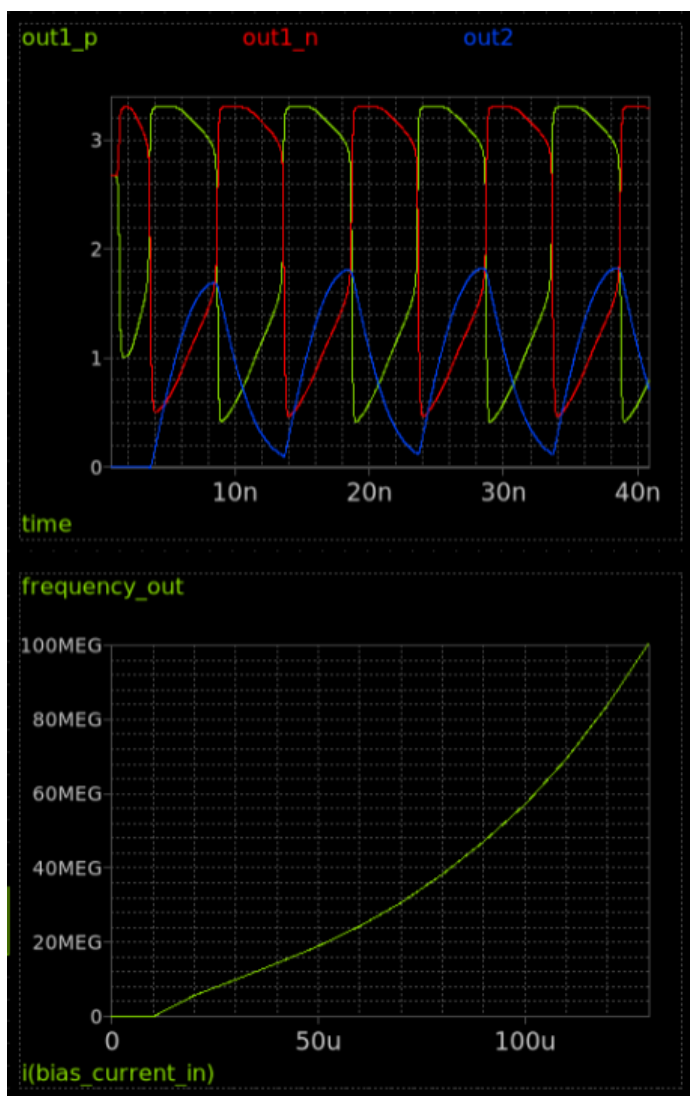
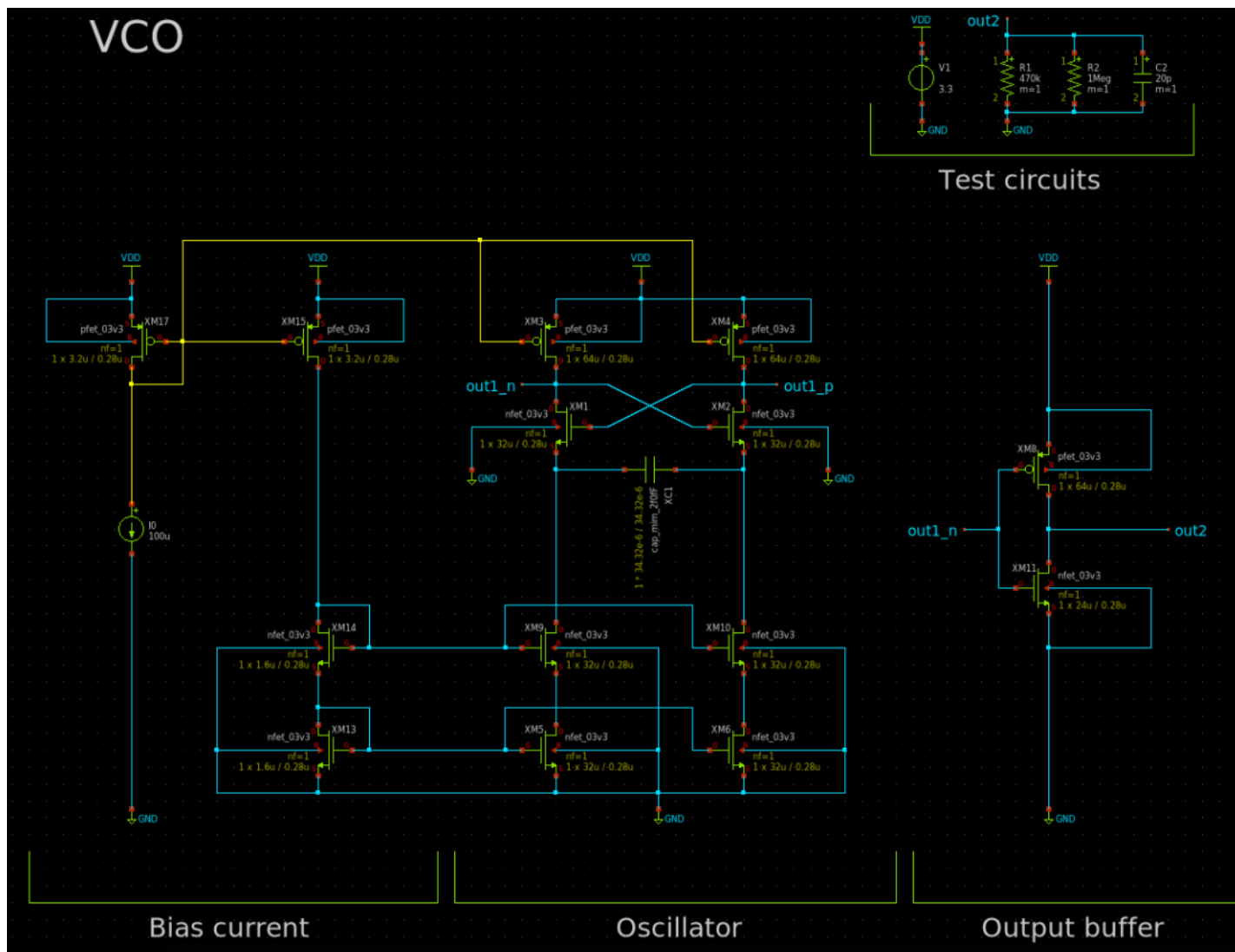
# Block diagram - Clock multiplier mode



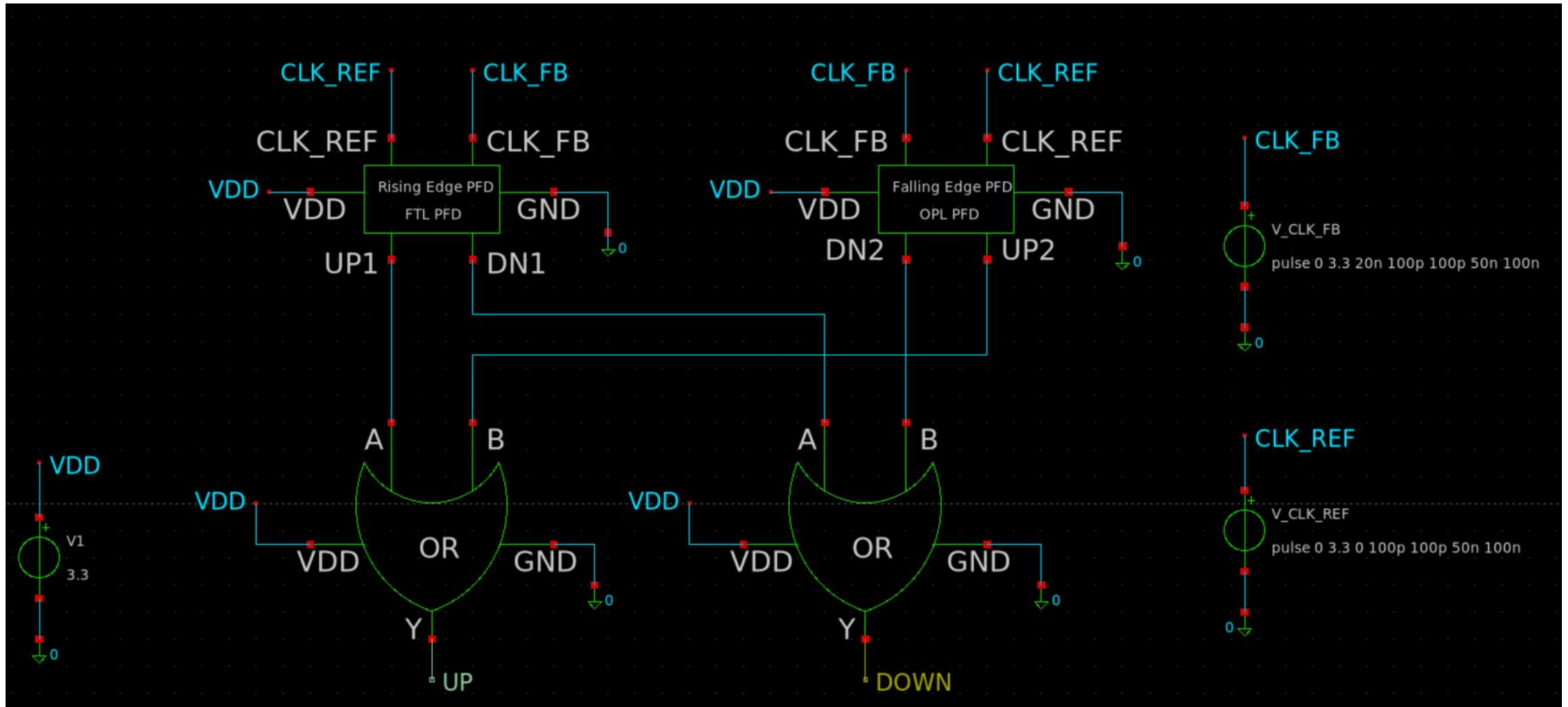
# Block diagram – FM demodulator mode



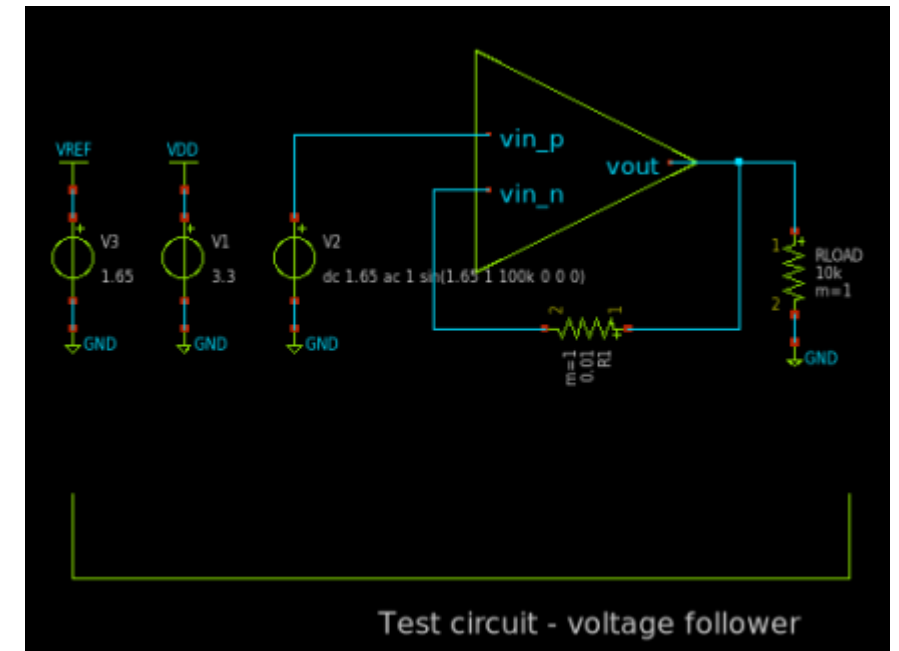
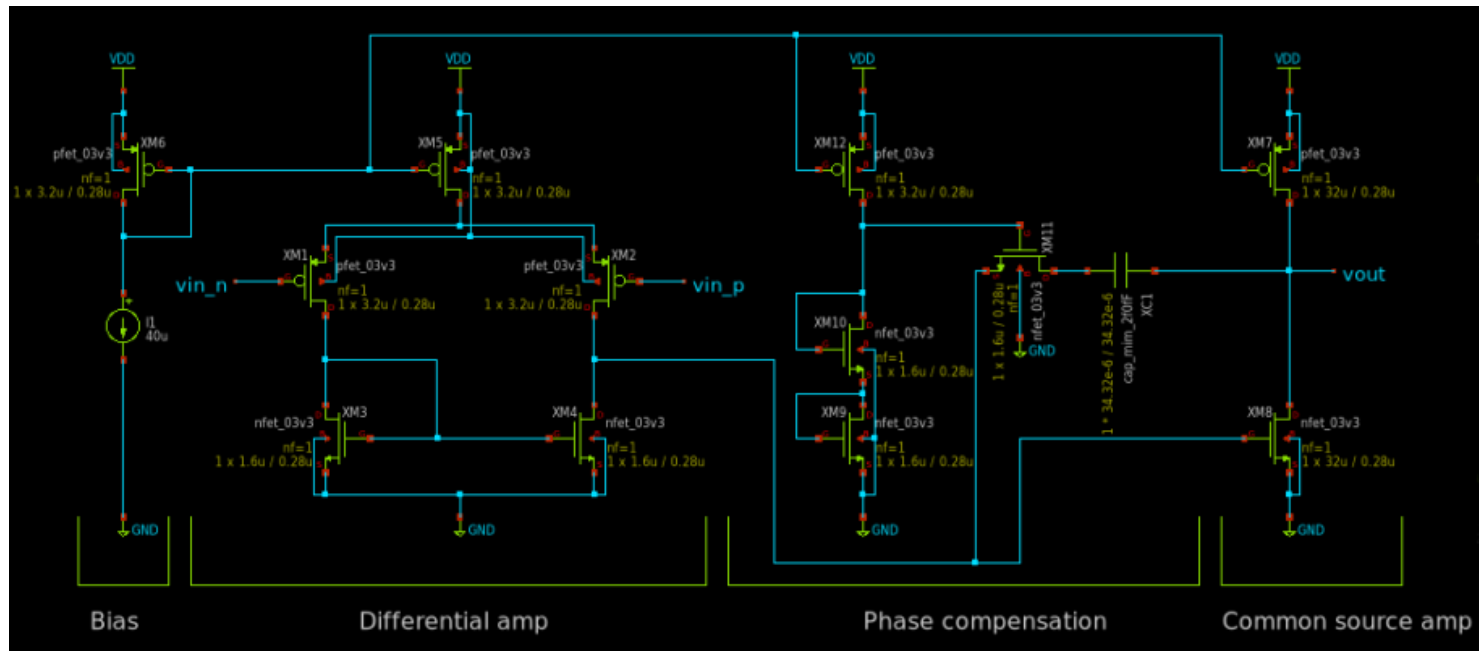
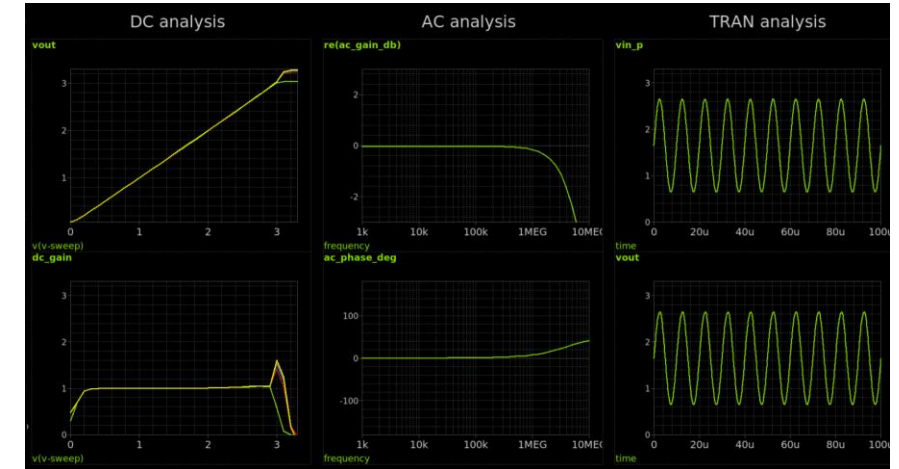
# Circuit design - VCO



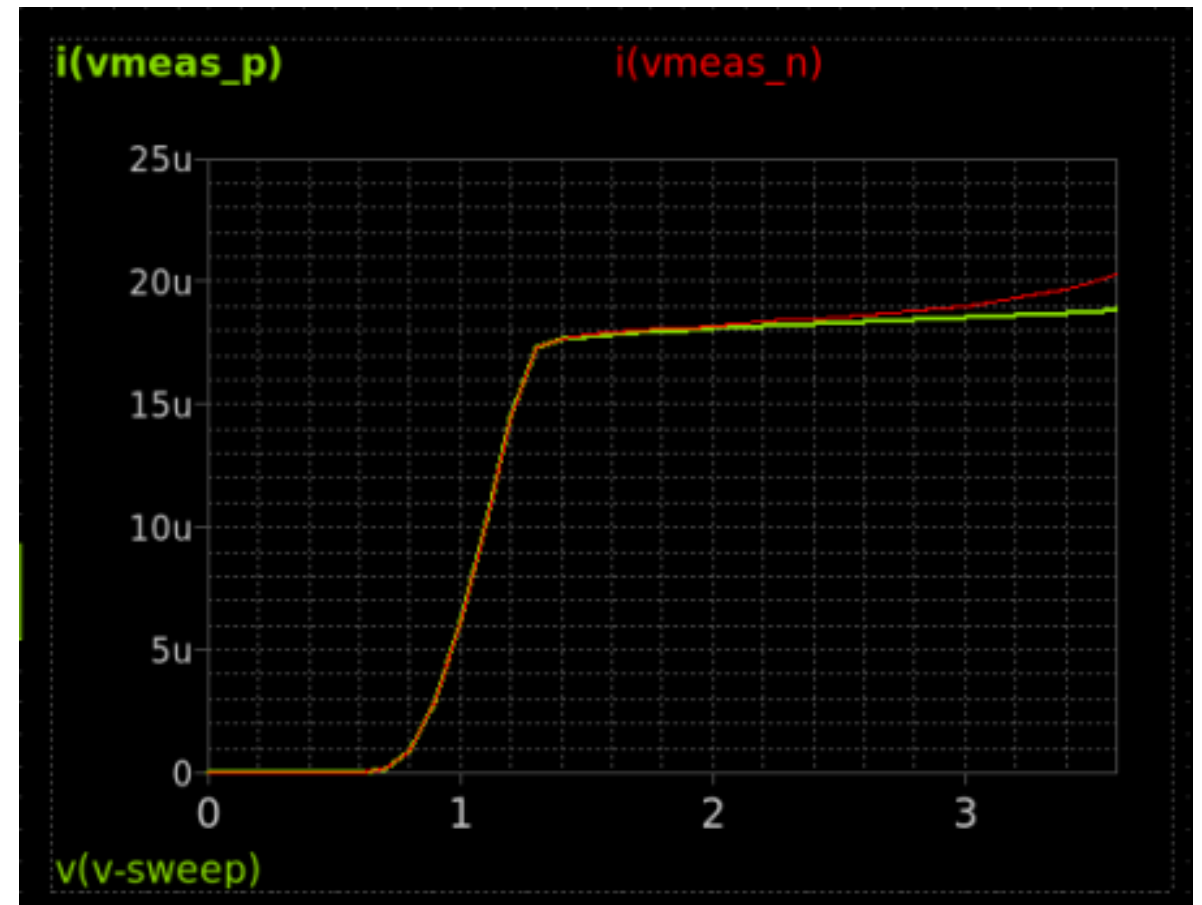
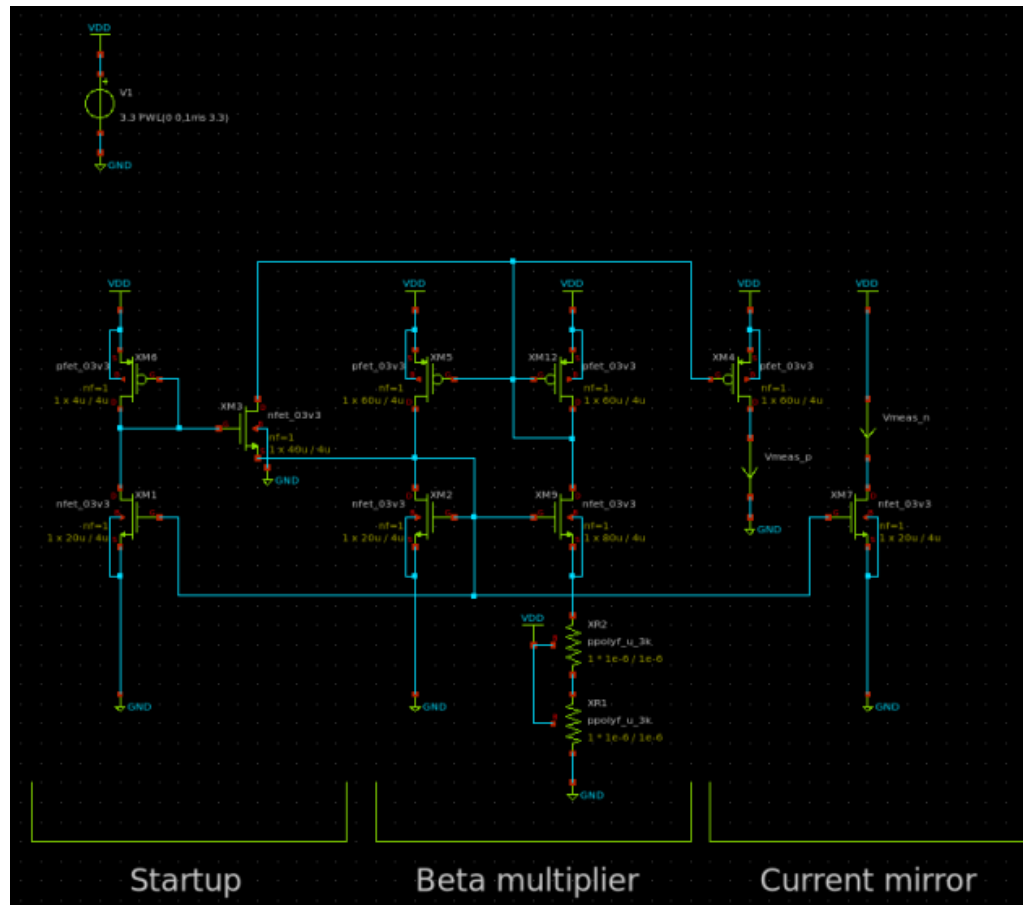
# Circuit design – PFD



# Circuit design - OPAMP



# Circuit design – Beta multiplier reference



# Layout (T.B.D)

# Evaluation (T.B.D)

- Put evaluation result here
- Insert graph to left

# Schedule

| Week    | Date    | Event                               | Track Details |
|---------|---------|-------------------------------------|---------------|
| Week 23 | June 5  | Team Formation Deadline             |               |
| Week 24 | June 12 | Project Proposal Review             |               |
| Week 27 | July 3  | Schematic Review                    |               |
| Week 28 | July 10 | Simulation Review (blocks)          |               |
| Week 29 | July 17 | Simulation Review (top level)       |               |
|         |         | READiness Check & Go/No-go Decision |               |
| Week 33 | Aug 14  | Layout Review (blocks)              |               |
| Week 34 | Aug 21  | Layout Review (top level)           |               |
| Week 35 | Aug. 28 | Verification                        |               |
|         |         | Final Chip Review                   |               |
|         |         | Final Submission                    |               |
|         |         | Upload of Chip Test Results         |               |